

Exploring the factors driving the success of bank telemarketing

Based on data from Portugese Banking Institution

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Introduction

In the competitive landscape of the banking industry, customer acquisition and retention are crucial for maintaining profitability and market share. One of the key strategies employed by banks is the promotion of term deposits, which are fixed-term investments offering a guaranteed return over a specified period. Understanding the factors that influence a client's decision to subscribe to a term deposit is essential for designing effective marketing campaigns. This analysis aims to investigate these factors using data from a Portuguese bank's marketing campaigns.

Background:

- Financial institutions often use marketing campaigns to promote their products.
- Understanding client characteristics and behaviors that lead to successful subscriptions can enhance campaign efficiency.

Key Terms:

- **Term Deposit:** A fixed-term investment where money is deposited for a specific duration with a guaranteed interest rate.
- **Marketing Campaign:** A series of coordinated activities and strategies to promote a product or service.

Problem Statement

- **Question:** What are the key demographic, financial, and behavioral factors that influence a client's decision to subscribe to a term deposit during a marketing campaign?
- **Approach:**
 - *Analyze client data including age, job, marital status, education, and financial indicators like balance, loan status, and previous campaign outcomes.*
 - *Use statistical and machine learning methods to identify patterns and significant predictors.*

Data

■ Data Source:

- The data used for this investigation comes from the UCI Machine Learning Repository, specifically the “Bank Marketing” dataset. It contains information on various client attributes and the results of a bank’s marketing campaign.

■ Sampling Method:

- The dataset includes data from direct marketing campaigns of a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact to the same client was required to ascertain if the product (bank term deposit) would be subscribed.

```
data_frame = read_csv2("bank-full.csv")  
  
head(data_frame)
```

Names of features available in data frame.

```
colnames(data_frame)
```

```
## [1] "age"      "job"      "marital"  "education" "default"  "balance"  
## [7] "housing"  "loan"     "contact"  "day"       "month"    "duration"  
## [13] "campaign" "pdays"   "previous" "outcome"   "y"
```

Variables:

- **age:** Age of the client (numeric)
- **job:** Type of job (categorical)
- **marital:** Marital status (categorical)
- **education:** Level of education (categorical)
- **default:** Has credit in default? (categorical: ‘yes’, ‘no’)
- **balance:** Average yearly balance in euros (numeric)
- **housing:** Has housing loan? (categorical: ‘yes’, ‘no’)
- **loan:** Has personal loan? (categorical: ‘yes’, ‘no’)
- **contact:** Contact communication type (categorical)
- **day:** Last contact day of the month (numeric)

- **month:** Last contact month of year (categorical)
- **duration:** Last contact duration in seconds (numeric)
- **campaign:** Number of contacts performed during this campaign (numeric)
- **pdays:** Number of days since the client was last contacted from a previous campaign (numeric, -1 means client was not previously contacted)
- **previous:** Number of contacts performed before this campaign (numeric)
- **poutcome:** Outcome of the previous marketing campaign (categorical)
- **y:** Has the client subscribed to a term deposit? (binary: 'yes', 'no')

Show the data types of each variable:

```
sapply(data_frame, class)
```

```
##      age      job      marital  education  default  balance
## "numeric" "character" "character" "character" "character" "numeric"
##      housing  loan      contact      day      month  duration
## "character" "character" "character" "numeric" "character" "numeric"
##      campaign  pdays  previous  poutcome      y
## "numeric"    "numeric" "numeric" "character" "character"
```

Data Pre-processing

- Load the data using **read_csv2** from the **readr** package, in previous steps.
- Handle missing values if any (e.g., by removing or imputing them).

```
# Check for missing values
sum(is.na(data_frame))
```

```
## [1] 0
```

- Convert categorical variables to factors for better handling in analysis.

```
# Convert categorical variables to factors
data_frame$job <- as.factor(data_frame$job)
data_frame$marital <- as.factor(data_frame$marital)
data_frame$education <- as.factor(data_frame$education)
data_frame$default <- as.factor(data_frame$default)
data_frame$housing <- as.factor(data_frame$housing)
data_frame$loan <- as.factor(data_frame$loan)
data_frame$contact <- as.factor(data_frame$contact)
data_frame$month <- as.factor(data_frame$month)
data_frame$poutcome <- as.factor(data_frame$poutcome)
data_frame$y <- as.factor(data_frame$y)
```

```
# Output
head(data_frame)
```


Descriptive Statistics and Visualization

To summarize and visualize the important variables in the data set, we will perform the following steps:

- 1. **Summary Statistics:** Generate summary statistics for key variables.
- 2. **Visualizations:** Create visualizations to highlight interesting features of the data.
- 3. **Handling Data Issues:** Address any missing data or outliers.

Summary Statistics

```
# Summary of the dataset
summary_data <- summary(data_frame)

# Display summary statistics
summary_data

##      age      job      marital      education
## Min.   :18.00  blue-collar:9732  divorced: 5207  primary   : 6851
## 1st Qu.:33.00  management :9458  married :27214  secondary:23202
## Median :39.00  technician :7597  single  :12790  tertiary :13301
## Mean   :40.94  admin.     :5171   unknown : 1857
## 3rd Qu.:48.00  services   :4154
## Max.   :95.00  retired    :2264
##              (Other) :6835
## default      balance      housing      loan      contact
## no :44396 Min.   : -8019 no :20081 no :37967 cellular :29285
## yes:  815 1st Qu.:   72  yes:25130 yes: 7244 telephone: 2906
##              Median :  448
##              Mean   : 1362
##              3rd Qu.: 1428
##              Max.   :102127
##
##      day      month      duration      campaign
## Min.   : 1.00  may :13766 Min.   : 0.0 Min.   : 1.000
## 1st Qu.: 8.00  jul : 6895 1st Qu.: 103.0 1st Qu.: 1.000
## Median :16.00  aug : 6247 Median : 180.0 Median : 2.000
## Mean   :15.81  jun : 5341 Mean   : 258.2 Mean   : 2.764
## 3rd Qu.:21.00  nov : 3970 3rd Qu.: 319.0 3rd Qu.: 3.000
## Max.   :31.00  apr : 2932 Max.   :4918.0 Max.   :63.000
##              (Other): 6060
##      pdays      previous      poutcome      y
## Min.   : -1.0 Min.   : 0.0000 failure: 4901 no :39922
## 1st Qu.: -1.0 1st Qu.: 0.0000 other : 1840 yes: 5289
## Median : -1.0 Median : 0.0000 success: 1511
## Mean   : 40.2 Mean   : 0.5803 unknown:36959
## 3rd Qu.: -1.0 3rd Qu.: 0.0000
## Max.   :871.0 Max.   :275.0000
##
```

Visualizations

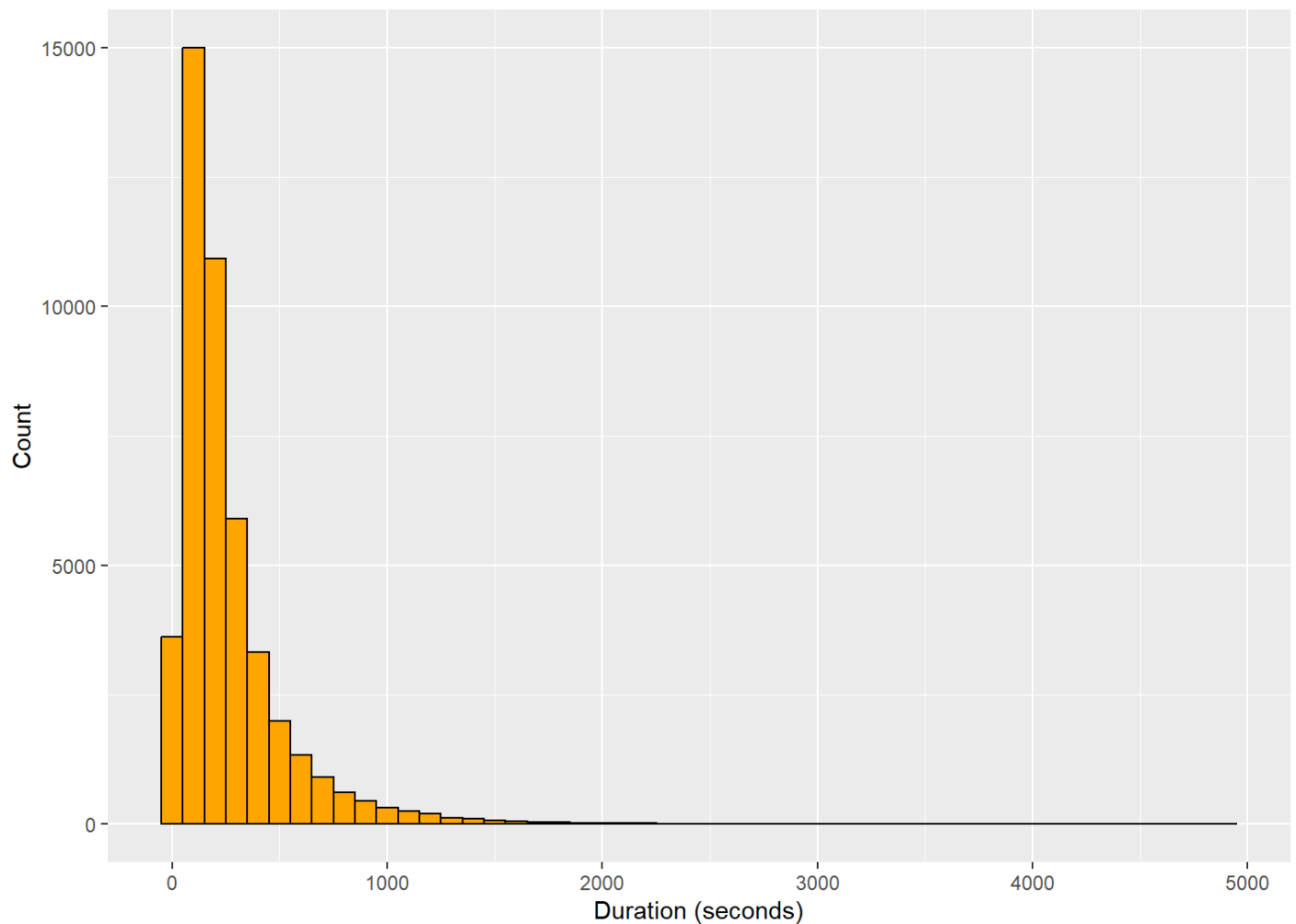
Duration of Last Contact

```
# Summary statistics for duration
summary(data_frame$duration)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0  103.0   180.0   258.2  319.0   4918.0

# Duration distribution plot
ggplot(data_frame, aes(x = duration)) +
  geom_histogram(binwidth = 100, fill = "orange", color = "black") +
  labs(title = "Distribution of Last Contact Duration", x = "Duration (seconds)", y = "Count")
```

Distribution of Last Contact Duration



Summary Statistics:

- The duration of the last contact ranges from 0 to 4918 seconds.
- The median duration is 180 seconds, meaning half of the calls last less than 3 minutes.
- The mean duration is 258.29 seconds, showing a right-skewed distribution.

Visualization:

- The histogram displays a right-skewed distribution with most calls being relatively short.
- The majority of the calls are under 600 seconds (10 minutes), with a few outliers.

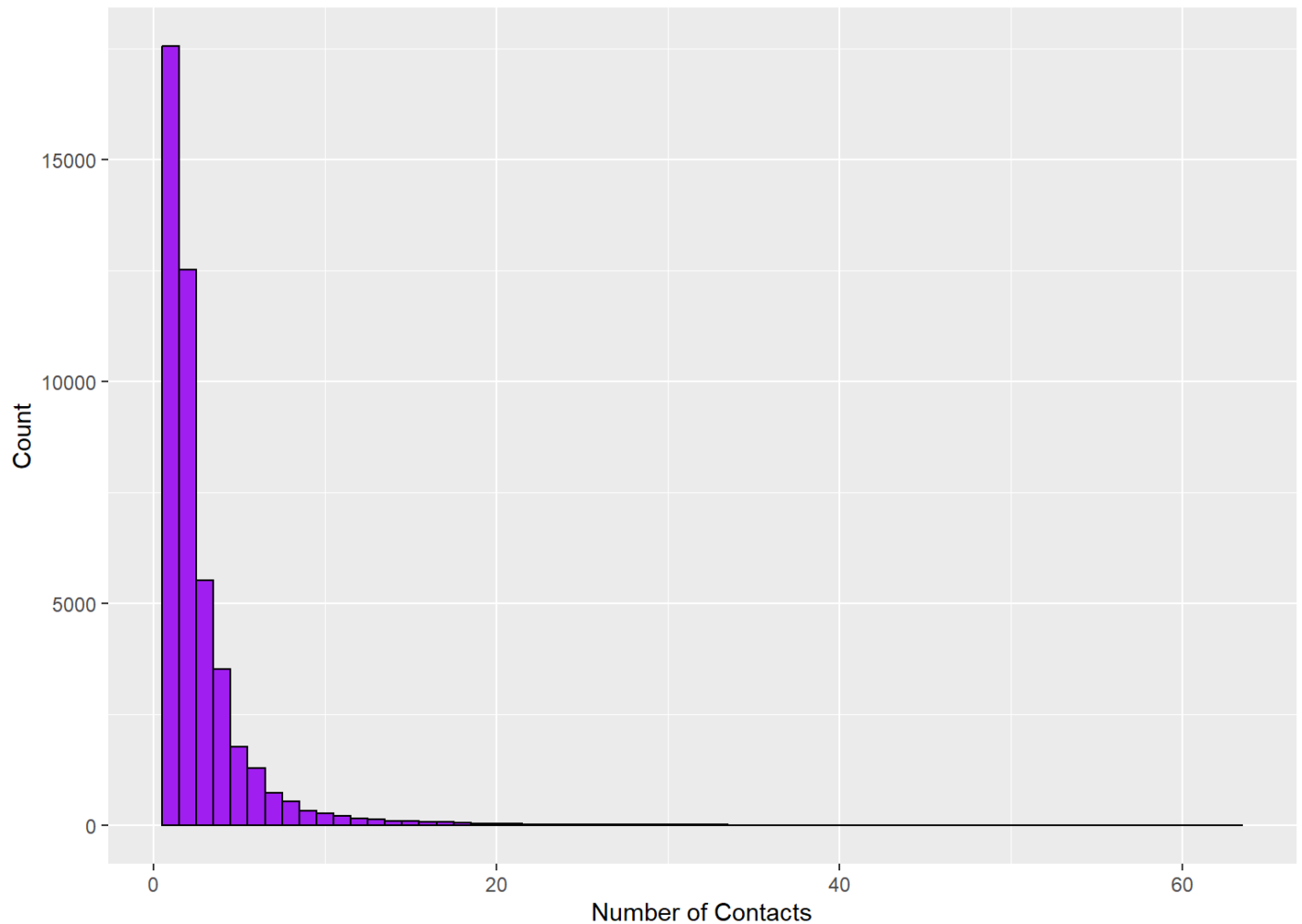
Number of Contacts During Campaign

```
# Summary statistics for campaign
summary(data_frame$campaign)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000  1.000   2.000   2.764  3.000   63.000
```

```
# Campaign contacts distribution plot
ggplot(data_frame, aes(x = campaign)) +
  geom_histogram(binwidth = 1, fill = "purple", color = "black") +
  labs(title = "Distribution of Number of Contacts During Campaign", x = "Number of Contacts", y = "Count")
```


Distribution of Number of Contacts During Campaign



Summary Statistics:

- The number of contacts ranges from 1 to 50.
- The median number of contacts is 2, with a mean of 2.764.
- Most clients are contacted between 1 and 3 times.

Visualization:

- The histogram shows that most clients receive 1 to 3 contacts.
- A few clients receive a significantly higher number of contacts, indicating persistent follow-up strategies.

Days Since Last Contact from Previous Campaign

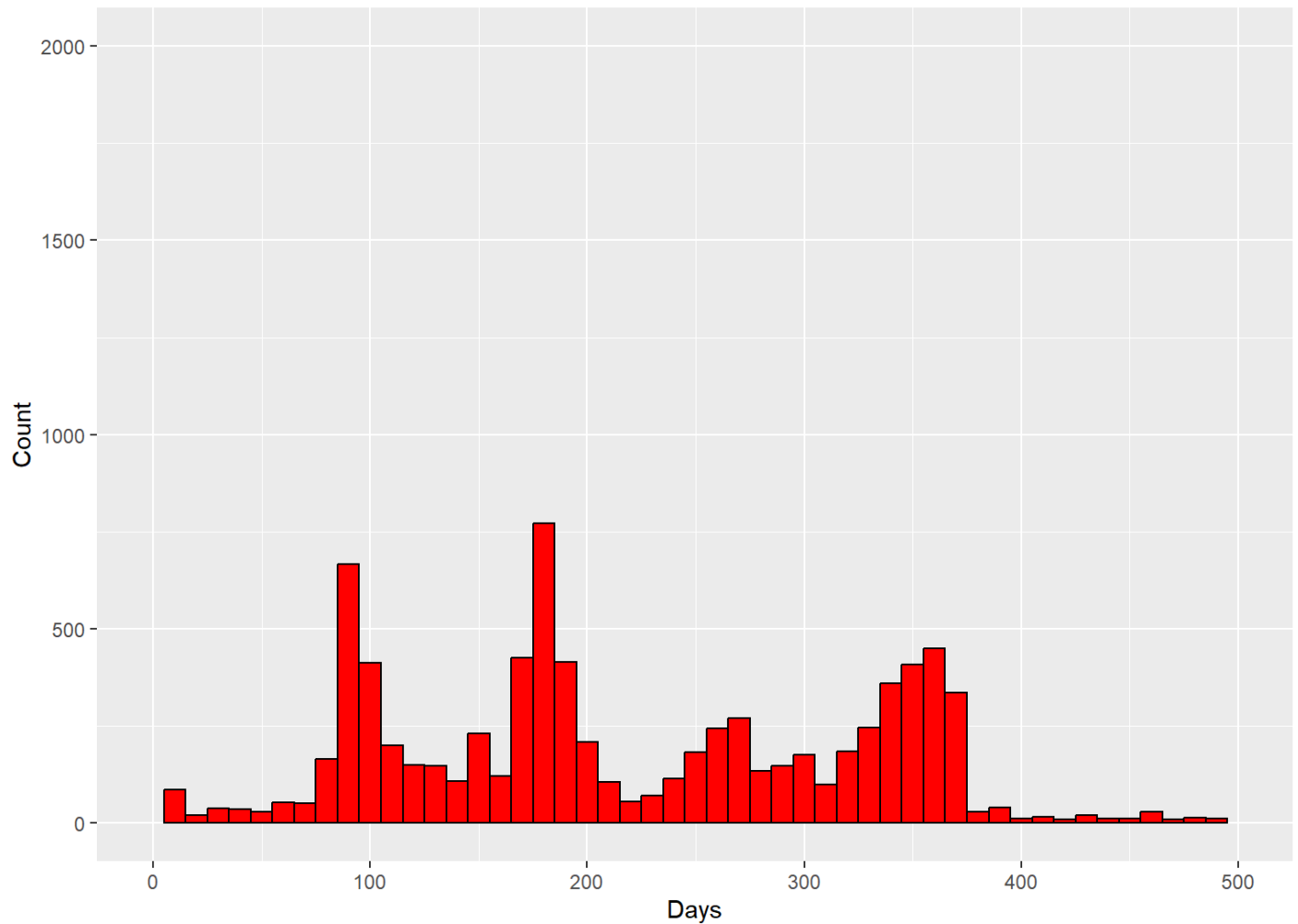
```
# Summary statistics for pdays
summary(data_frame$pdays)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	-1.0	-1.0	-1.0	40.2	-1.0	871.0

```
# Pdays distribution plot
ggplot(data_frame, aes(x = pdays)) +
  geom_histogram(binwidth = 10, fill = "red", color = "black") +
  labs(title = "Distribution of Days Since Last Contact from Previous Campaign", x = "Days", y = "Count") +
```

```
xlim(-1, 500)+ # Limiting x-axis for better visualization
ylim(0,2000)
```

Distribution of Days Since Last Contact from Previous Campaign



Summary Statistics:

- Many clients have -1 days since the last contact, indicating no previous contact.
- For those with previous contacts, days range from 0 to 871.
- The mean number of days is 40.2, indicating infrequent previous contacts.

Visualization:

- The histogram shows a large spike at -1, indicating many clients had no previous contact.
- Other values are spread out, showing the infrequency and wide range of past contacts.

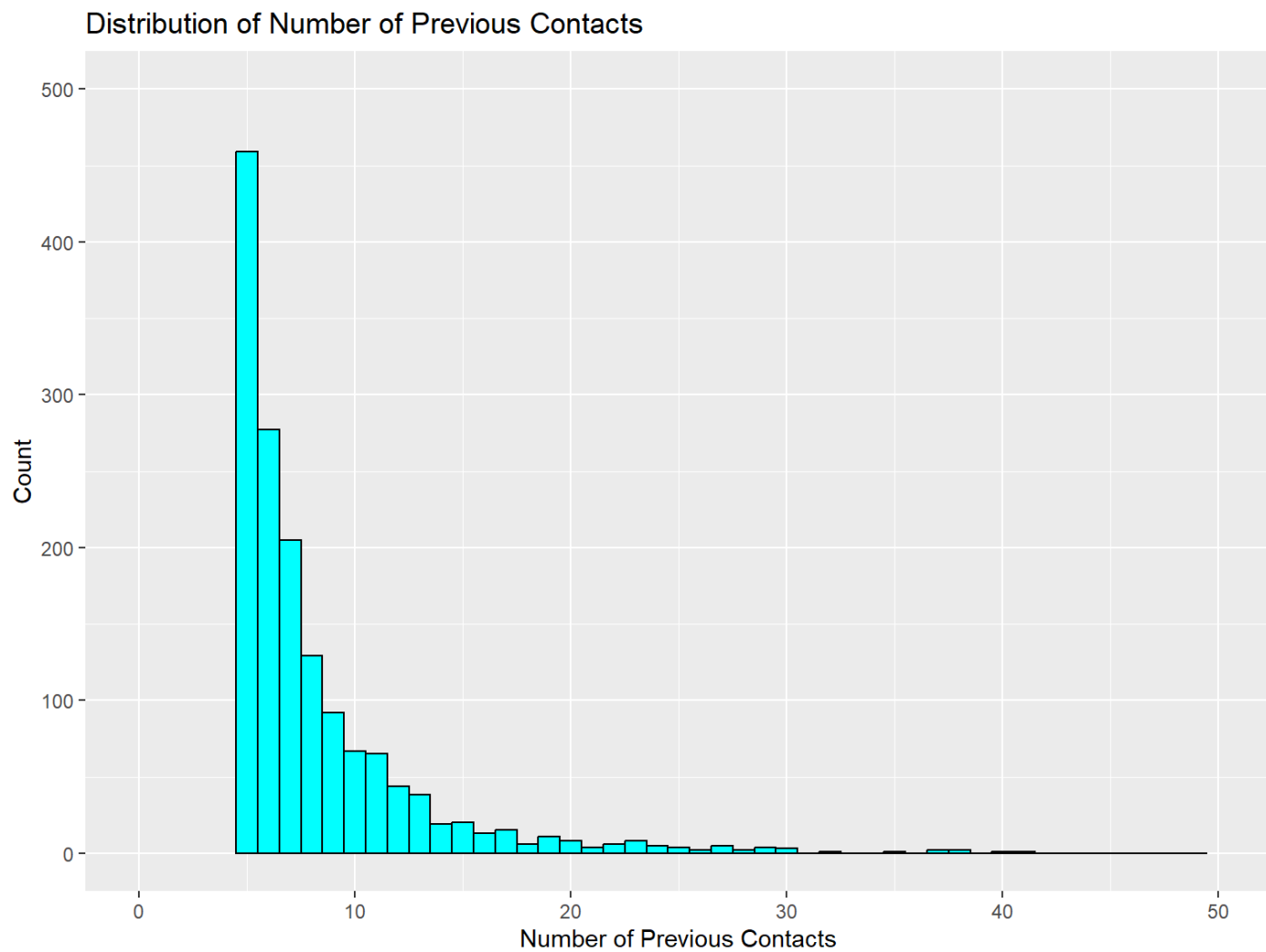
Previous Campaign Outcomes

```
# Summary statistics for previous
summary(data_frame$previous)
```

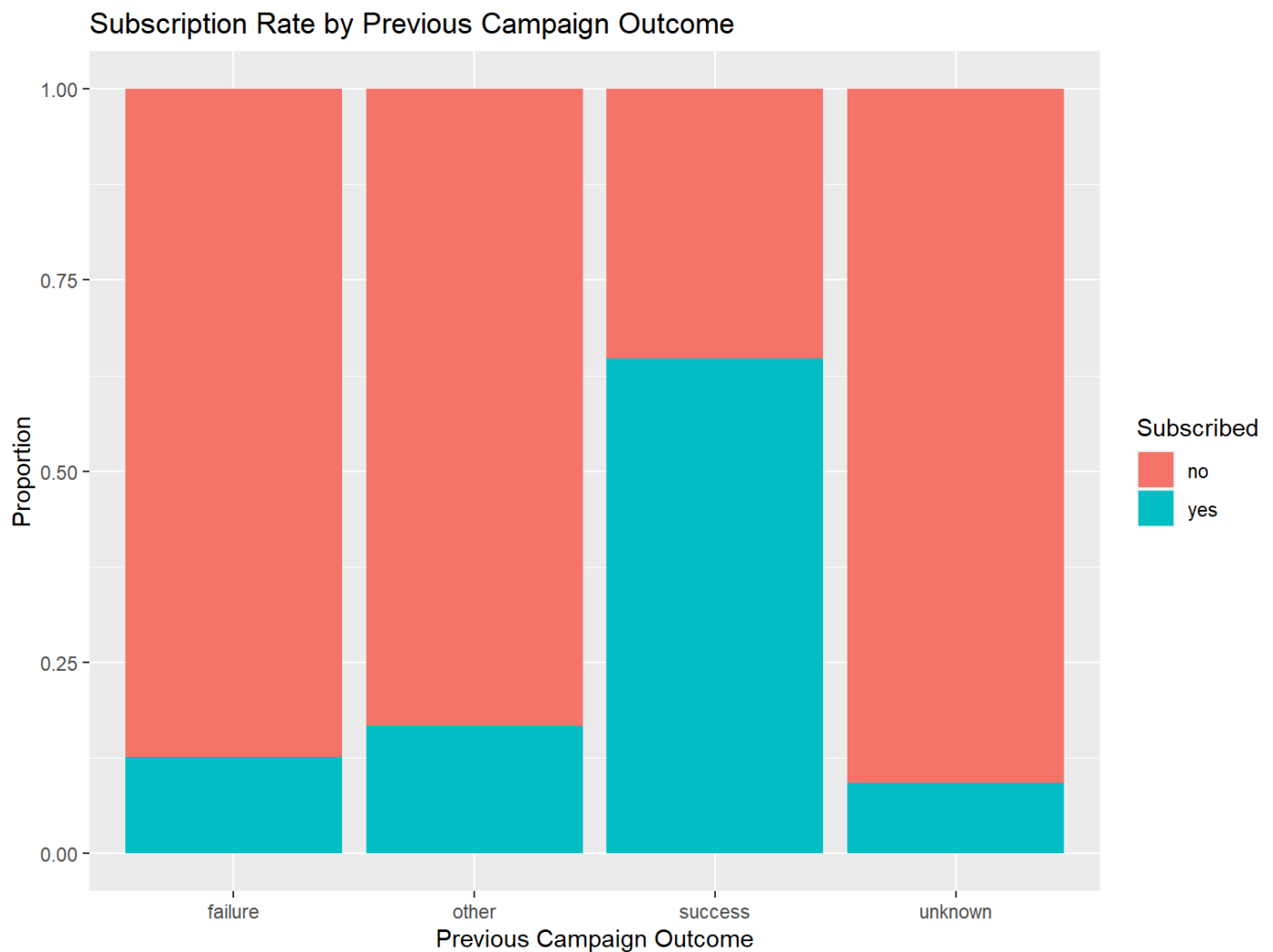
```
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
##  0.0000   0.0000   0.0000   0.5803   0.0000  275.0000
```

```
# Previous contacts distribution plot
ggplot(data_frame, aes(x = previous)) +
```

```
geom_histogram(binwidth = 1, fill = "cyan", color = "black") +
labs(title = "Distribution of Number of Previous Contacts", x = "Number of Previous Contacts", y = "Count")+
xlim(0,50)+
ylim(0,500)
```



```
# Subscription rate by previous campaign outcome
ggplot(data_frame, aes(x = poutcome, fill = y)) +
  geom_bar(position = "fill") +
  labs(title = "Subscription Rate by Previous Campaign Outcome", x = "Previous Campaign Outcome", y = "Proportion", fill = "Subscribed")
```



Summary Statistics:

- Most clients had zero previous contacts, with a few having multiple past contacts.
- The median and 3rd quartile values are 0, highlighting that previous contacts were rare.

Visualization:

- The histogram shows a sharp drop after zero, indicating most clients had no previous contacts.
- The bar plot shows clients with a successful previous outcome (**poutcome**) are more likely to subscribe again, highlighting the importance of past campaign success.

By examining the duration of last contact, number of campaign contacts, days since last contact, and previous campaign outcomes, we can better understand client behaviors and improve marketing strategies. The data indicates that previous successful contacts and specific client demographics significantly influence subscription decisions. These findings will help tailor marketing efforts to target potential clients more effectively.

Hypothesis testing

For this task, we will test if there is a significant difference in the balance of clients who subscribed to a term deposit ($y = \text{yes}$) and those who did not ($y = \text{no}$). We will also calculate a confidence interval for the mean difference in balance between these two groups.

Hypotheses: We want to test if there is a significant difference in the mean balance between the two groups.

Checking Assumptions

1. **Normality of Balance for Both Groups:** We will use the Shapiro-Wilk test to check for normality.
2. **Equal Variances:** We will use Levene's test to check for equal variances between the two groups.

Since the balance data might not be normally distributed, we will use the Wilcoxon rank-sum test (a non-parametric test) if the normality assumption is violated.

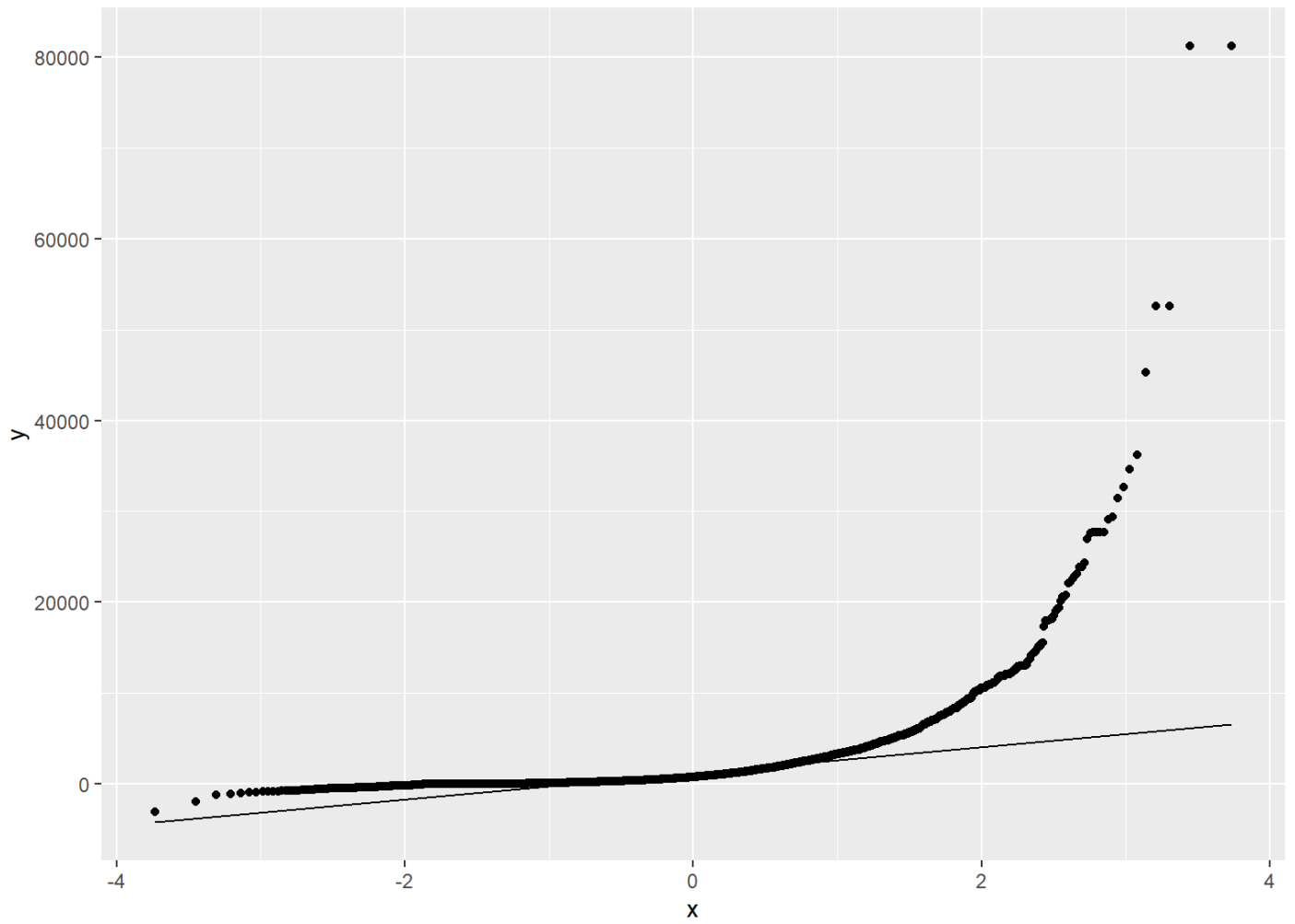
```
# Filter the dataset for the two groups
subscribed <- data_frame %>% filter(y == "yes")
not_subscribed <- data_frame %>% filter(y == "no")
```

```
# Visual inspection using Q-Q plots
qqplot_yes <- ggplot(subscribed, aes(sample = balance)) +
  stat_qq() + stat_qq_line() +
  ggtitle("Q-Q Plot for Subscribed Group")

qqplot_no <- ggplot(not_subscribed, aes(sample = balance)) +
  stat_qq() + stat_qq_line() +
  ggtitle("Q-Q Plot for Not Subscribed Group")

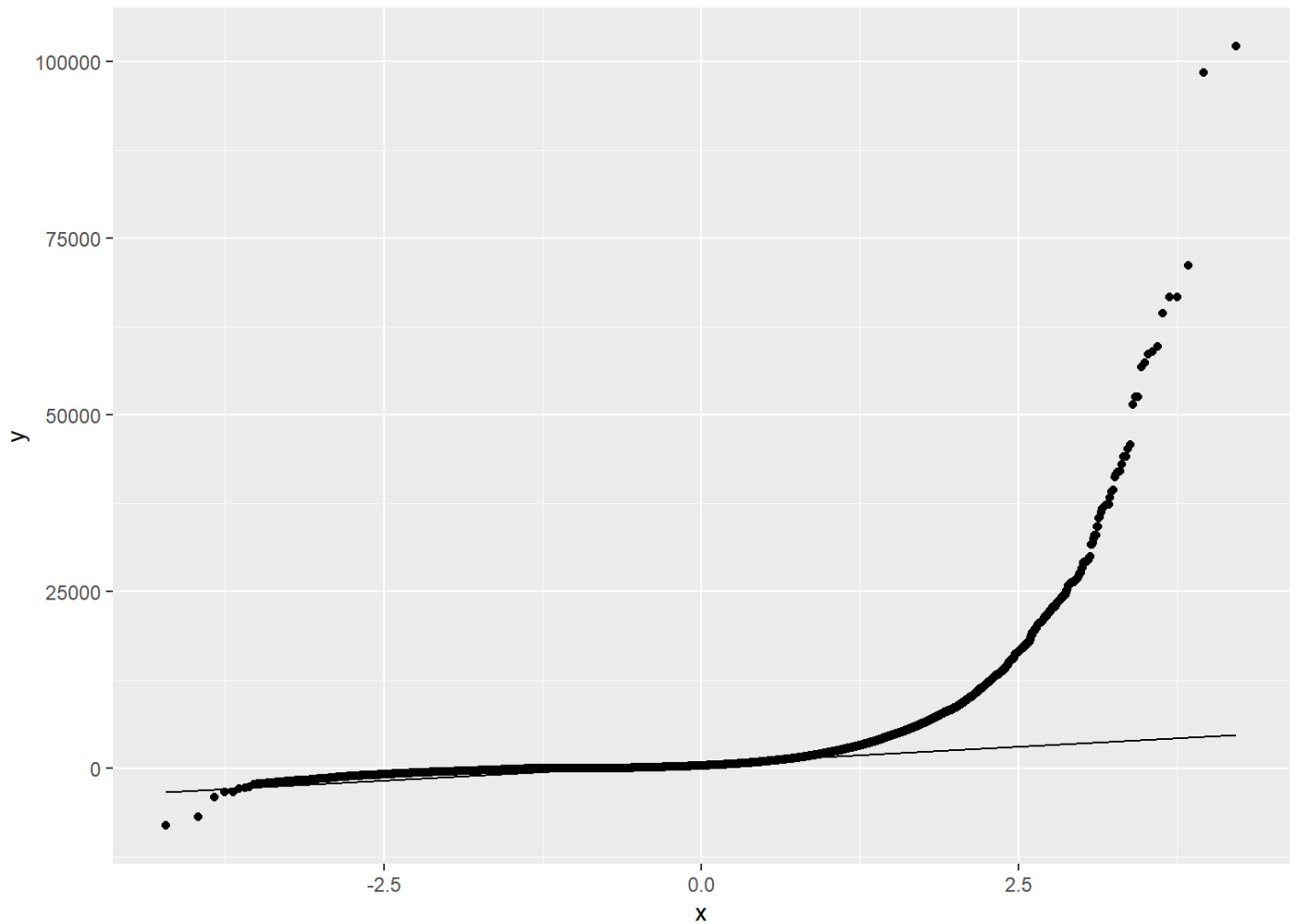
# Print the Q-Q plots to display them
print(qqplot_yes)
```

Q-Q Plot for Subscribed Group



```
print(qqplot_no)
```

Q-Q Plot for Not Subscribed Group



```
# Check normality using Kolmogorov-Smirnov test
ks_test_yes <- ks.test(subscribed$balance, "pnorm", mean(subscribed$balance), sd(subscribed$balance))
ks_test_no <- ks.test(not_subscribed$balance, "pnorm", mean(not_subscribed$balance), sd(not_subscribed$balance))

# Check for equal variances using Levene's test
levene_test <- leveneTest(balance ~ y, data = data_frame)

# Perform Wilcoxon rank-sum test (non-parametric) if normality assumption is violated
wilcox_test <- wilcox.test(balance ~ y, data = data_frame)

# Calculate the confidence interval for the mean difference in balance
mean_diff <- mean(subscribed$balance) - mean(not_subscribed$balance)
std_error <- sqrt(var(subscribed$balance)/nrow(subscribed) + var(not_subscribed$balance)/nrow(not_subscribed))
conf_interval <- mean_diff + c(-1, 1) * qnorm(0.975) * std_error

# Output the results
list(
  ks_test_yes = ks_test_yes,
  ks_test_no = ks_test_no,
  levene_test = levene_test,
  wilcox_test = wilcox_test,
  mean_difference = mean_diff,
  confidence_interval = conf_interval
)
```

```
## $ks_test_yes
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: subscribed$balance
## D = 0.2672, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
## $ks_test_no
##
## Asymptotic one-sample Kolmogorov-Smirnov test
```

```
##
## data: not_subscribed$balance
## D = 0.26042, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
##
## $levene_test
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group  1  61.295 5.02e-15 ***
##      45209
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $wilcox_test
##
## Wilcoxon rank sum test with continuity correction
##
## data: balance by y
## W = 86557475, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
##
## $mean_difference
## [1] 500.5529
##
## $confidence_interval
## [1] 401.7900 599.3158
```

1. Normality and Equal Variances:

- If the Q-Q plots show significant deviations from the normal line, it indicates non-normality.
- If the p-values from the Kolmogorov-Smirnov tests are less than 0.05, we reject the null hypothesis of normality.
- If the p-value from Levene's test is less than 0.05, we reject the null hypothesis of equal variances.

2. Wilcoxon Rank-Sum Test:

- **wilcox_test:** This test compares the median balance between the subscribed and non-subscribed groups. The test statistic W is 86557475, and the p-value is $< 2.2e-16$, indicating a significant difference in the median balance between the two groups.

3. Mean Difference and Confidence Interval:

- **mean_difference:** The mean difference in balance between subscribed and non-subscribed clients is 500.5529 units.
- **confidence_interval:** The confidence interval for the mean difference is [401.7900, 599.3158], which means we are 95% confident that the true mean difference falls within this range.

Regression Analysis

For this task, we will perform a logistic regression analysis to identify the key demographic, financial, and behavioral predictors of whether a client subscribes to a term deposit (y).

```
# Fit a logistic regression model
logistic_model <- glm(y ~ age + job + marital + education + balance + housing + loan + previous + poutcome,
  data = data_frame, family = binomial)

# Summarize the model
summary(logistic_model)
```

```
##
## Call:
## glm(formula = y ~ age + job + marital + education + balance +
##      housing + loan + previous + poutcome, family = binomial,
##      data = data_frame)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.639e+00  1.324e-01 -12.383  < 2e-16 ***
## age           1.300e-03  1.902e-03   0.684  0.494152
## jobblue-collar -2.506e-01  6.336e-02  -3.956  7.64e-05 ***
```



```
## jobentrepreneur -3.299e-01 1.096e-01 -3.012 0.002599 **
## jobhousemaid -3.700e-01 1.174e-01 -3.153 0.001616 **
## jobmanagement -1.209e-01 6.397e-02 -1.891 0.058674 .
## jobretired 5.161e-01 8.263e-02 6.246 4.21e-10 ***
## jobself-employed -1.962e-01 9.704e-02 -2.021 0.043236 *
## jobservices -1.826e-01 7.279e-02 -2.508 0.012129 *
## jobstudent 4.710e-01 9.605e-02 4.904 9.38e-07 ***
## jobtechnician -1.704e-01 6.000e-02 -2.839 0.004523 **
## jobunemployed 1.085e-01 9.508e-02 1.141 0.254004 .
## jobunknown -3.436e-01 2.015e-01 -1.705 0.088106 .
## maritalmarried -1.832e-01 5.045e-02 -3.630 0.000283 ***
## maritalsingle 1.692e-01 5.740e-02 2.947 0.003210 **
## educationsecondary 1.710e-01 5.552e-02 3.079 0.002076 **
## educationtertiary 4.004e-01 6.454e-02 6.204 5.52e-10 ***
## educationunknown 2.063e-01 8.958e-02 2.303 0.021289 *
## balance 2.170e-05 4.202e-06 5.164 2.42e-07 ***
## housingyes -6.934e-01 3.379e-02 -20.519 < 2e-16 ***
## loanyes -4.638e-01 5.153e-02 -9.001 < 2e-16 ***
## previous 7.468e-03 5.695e-03 1.311 0.189772
## poutcomeother 2.518e-01 7.813e-02 3.223 0.001267 **
## poutcomesuccess 2.220e+00 7.124e-02 31.160 < 2e-16 ***
## poutcomeunknown -4.296e-01 5.083e-02 -8.453 < 2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 32631 on 45210 degrees of freedom
## Residual deviance: 28720 on 45186 degrees of freedom
## AIC: 28770
##
## Number of Fisher Scoring iterations: 5
```

Regression Analysis

Model: A logistic regression model was used to identify the significant predictors of whether a client subscribes to a term deposit or not.

Key Findings from the Logistic Regression:

■ Demographic Factors:

- **Job:** Certain job types were significant predictors. For example, retired and student clients were more likely to subscribe, whereas blue-collar, entrepreneur, and housemaid clients were less likely.
- **Marital Status:** Married clients were less likely to subscribe, while single clients were more likely.
- **Education:** Higher education levels (secondary and tertiary) increased the likelihood of subscribing.

■ Financial Factors:

- **Balance:** Higher balances were associated with a higher likelihood of subscribing.
- **Housing Loan:** Clients with housing loans were less likely to subscribe.
- **Personal Loan:** Clients with personal loans were less likely to subscribe.

■ Behavioral Factors:

- **Previous Campaign Outcome:** Clients with a successful previous campaign outcome were significantly more likely to subscribe.
- **Previous Contacts:** Although not significant, the number of previous contacts showed a positive trend.

Discussion

- The investigation revealed significant demographic, financial, and behavioral predictors for term deposit subscription.
- Higher balances and previous successful campaign outcomes positively influenced the likelihood of subscribing, while having housing or personal loans decreased it.
- Certain job types (e.g., retired and student) and higher education levels also increased subscription rates. Hypothesis testing confirmed a significant difference in balances between subscribers and non-subscribers.
- The **strengths** of the analysis include a comprehensive data set and robust statistical methods, while **limitations** involve potential biases and unaccounted external factors. Future research should incorporate additional variables and explore advanced machine learning models for improved predictions.

References

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